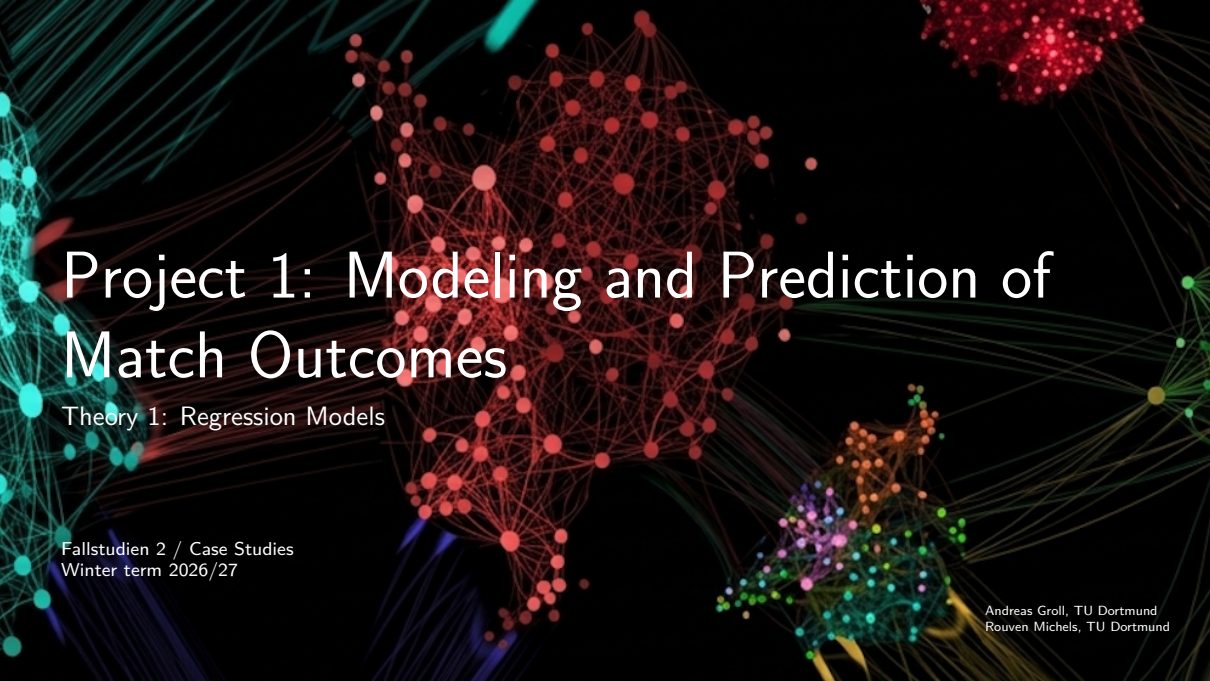


Project 1: Modeling and Prediction of Match Outcomes

Theory 1: Regression Models

Fallstudien 2 / Case Studies
Winter term 2026/27

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Introduction

Introduction

- Lecturers: **Andreas Groll** (groll@statistik.tu-dortmund.de) and **Rouven Michels** (rouven.michels@tu-dortmund.de)
- Special thanks for helping to prepare the course materials goes to **Dr. Gunther Schauberger** (TU Munich)
- This project is dedicated to the modeling of match outcomes
- Match outcomes occur whenever two competitors face each other in a match
- We will present different strategies/approaches to model the most spread types of match outcomes and illustrate these using real-world data from football, tennis, and handball
- Beside classic regression models, we will also give some insights into more advanced machine learning approaches

Outline

- Project 1 consists of theory units and lab sessions.
- Theory topics
  - Theory 1: Introduction to regression models
  - Theory 2: Introduction to Regression and Classification Trees
  - Theory 3: Predictive modeling and Random Forests
- We will decide flexibly on the exact order between theory and lab parts.

Match Outcomes

Match Outcomes

- The **match outcome** variable can vary substantially between different kinds of sports.
- Even for the same sport, we may have various possibilities to define the outcome variable.
- Mathematically, match outcomes can be **binary, ordinal, or metric variables**.
- Depending on the variable type, different frameworks may be suitable to model these outcomes.

Our main focus in this first session will be on **regression models** for different types of match outcomes. In the next sessions, two machine learning methods will be introduced and applied to modeling match outcomes.

Match Outcomes (Examples)

- Binary match outcome
  - Tennis (1: Player A wins, 0: Player B wins)
  - Basketball (1: Team A wins, 0: Team B wins)
- Ordinal match outcome
  - Football (1: Team A wins, 2: Draw, 3: Team B wins)
  - Football (1: Team A wins with more than 1 goal diff., 2: Team A wins with 1 goal diff., 3: Draw, 4: Team B wins with 1 goal diff., 5: Team B wins with more than 1 goal diff.)
- Metric match outcome
  - Football, Handball, Basketball: number of scores / goals per team (i.e. two observations per match)
  - Football, Handball, Basketball: score difference between teams A and B (i.e. one observation per match)

Regression for Match Outcomes

- For different types of outcomes, different types of (regression) models need to be used.
  - Binary outcome: logistic regression
  - Ordinal outcome: ordinal regression
  - Metric Outcome:
    - Linear regression (assuming normal distribution)
    - Poisson regression (assuming a Poisson distribution)
    - Skellam regression (for score differences, if normal distribution does not fit)

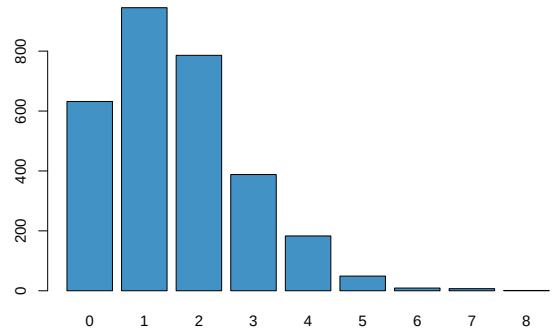
$\Rightarrow$ We'll try to give some hints on when to choose which type of regression model.

Scores as Match Outcomes

- We first consider **scores** (e.g. number of goals) as the match outcome, which is a **metric variable**.
- In this case, we have **two observations per match**.
  - We assume independence between the two scores from one match (conditioning on the predictors)
  - If this is no viable assumption, we need a way to account for the dependence (not considered today).
- The most plausible options for modeling are
  - **Linear regression** (for rather large scores, e.g. Basketball or Handball)
    - Simplest solution, but scores are discrete variables while the normal distribution is a continuous distribution.
    - For small scores, the assumption of a normal distribution is not justified and allows for negative predictions (which is implausible for match outcomes).
  - **Poisson regression** (for small score, e.g. Football)
    - If the normal distribution is questionable for a metric match outcome, the Poisson distribution is the most natural alternative.

Scores as Match Outcomes

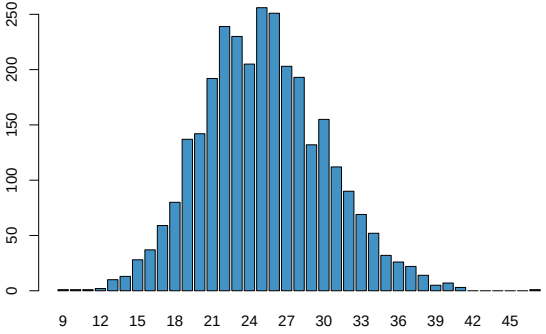
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- Exemplary distribution of goals per team in **football**
- **Not approximable** by a normal distribution

Scores as Match Outcomes

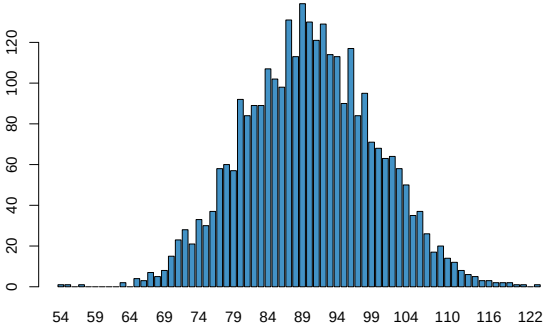
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- Exemplary distribution of goals per team in **handball**
- **Quite well approximable** by a normal distribution

Scores as Match Outcomes

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- Exemplary distribution of scores per team in **basketball**
- **Perfectly approximable** by a normal distribution

Metric Match Outcomes

- Now we consider **score differences** (e.g. goal difference) as the match outcome, which is also a **metric outcome variable**.
- In this case, we have 1 **observation per match**.
- Score difference can be both negative and positive. However, for both cases they are discrete while the normal distribution is continuous.
- Therefore, again, the normal distribution should only be used if there is a rather wide range of possible values (for rather large scores).
- If the normal distribution is not justified, we can use the Skellam distribution as an alternative.
- Technically, the Skellam distribution represents the distribution of the difference between two (independent) Poisson distributions!

Metric Match Outcomes

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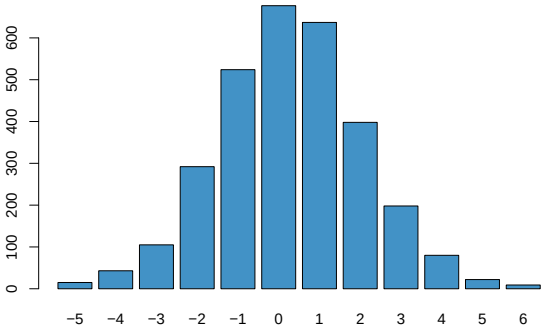
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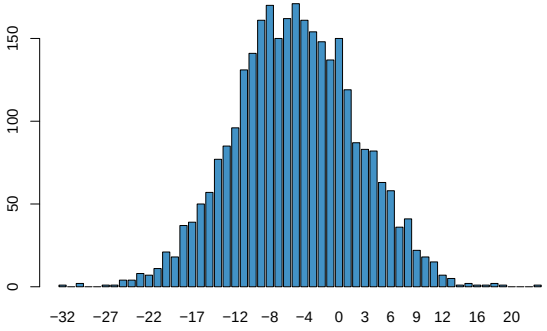
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- Exemplary distribution of goal difference in **football**
- **Not approximable** by a normal distribution

Metric Match Outcomes

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- Exemplary distribution of goal difference in **handball**
- **Quite well approximable** by a normal distribution

Metric Match Outcomes

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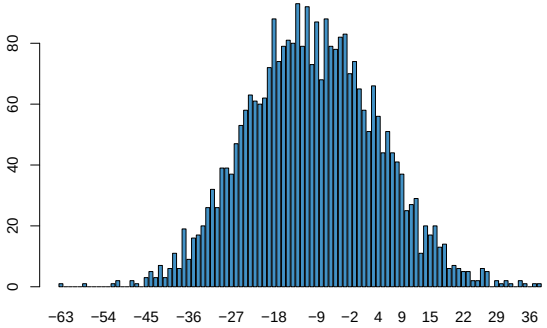
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- Exemplary distribution of score difference in **basketball**
- **Perfectly approximable** by a normal distribution

1

Exemplary Datasets

Handball Data

- The handball data stem from the first division of handball (Handball Bundesliga) for males in Germany; they contain 1,760 matches from the six seasons 2017/18 until 2022/23.
- The data are prepared in two versions:
  - match outcome: absolute number of goals (`handball_long`)
  - match outcome: goal difference (`handball_wide`)
- The data contain the following variables:
  - goals
  - season
  - bet.probs: winning probability according to bookmakers' odds
  - home_away: indicator for home team and away team

Handball Data

```
head(handball_long)
```

```
##   matchid                team season goals home_away bet.probs
## 1       1    Eulen Ludwigshafen   2017    25      home  25.86746
## 2       1 FRISCH AUF! Göppingen   2017    32      away  63.86586
## 3       2    Eulen Ludwigshafen   2017    19      home  12.62498
## 4       2         Füchse Berlin   2017    25      away  81.01946
## 5       3    Eulen Ludwigshafen   2017    21      home  53.98379
## 6       3            GWD Minden   2017    20      away  34.26278
```

```
head(handball_wide)
```

```
##   matchid          hometeam             awayteam season homegoals awaygoals scorediff bet.probs.diff
## 1       1 Eulen Ludwigshafen FRISCH AUF! Göppingen   2017        25        32        -7     -0.3799840
## 2       2 Eulen Ludwigshafen         Füchse Berlin   2017        19        25        -6     -0.6839448
## 3       3 Eulen Ludwigshafen           GWD Minden   2017        21        20         1      0.1972102
## 4       4 Eulen Ludwigshafen     Hannover-Burgdorf   2017        21        27        -6     -0.4803739
## 5       5 Eulen Ludwigshafen           HC Erlangen   2017        32        29         3      0.3320547
## 6       6 Eulen Ludwigshafen           HSG Wetzlar   2017        31        29         2     -0.4484281
```

UEFA Euro Data

- The UEFA Euro data are separated into training data and test data.
  - The training data contain data from the UEFA Euros 2008 to 2020.
  - The test data contain data from the UEFA Euro 2024.
- The data contain the following predictors:
  - GroupStage: 1 for group stage, 0 for knockout stage
  - GDP: GDP of respective country
  - MarketValue: cumulative market value of squad
  - FifaRank: FIFA rank before the tournament
  - UefaPoints: number of UEFA points before tournament
  - CLPlayers: number of players participating in UEFA CL semifinal prior to tournament
- For modeling purposes, it can be advantageous to use all predictors in the form differences between team and opponent.

UEFA Euro Data

```
head(Euro_train)
```

```
##   Goals           Team       Opponent Year GroupStage        GDP MarketValue FifaRank UefaPoints CLPlayers
## 1     0    Switzerland Czech Republic 2008          1  1.1550862  -0.3133334       42      -10700        -1
## 2     1 Czech Republic    Switzerland 2008          1 -1.1550862   0.3133334      -42       10700         1
## 3     2       Portugal         Turkey 2008          1  0.8342002   0.9367693      -16       10042         5
## 4     0         Turkey       Portugal 2008          1 -0.8342002  -0.9367693       16      -10042        -5
## 5     1 Czech Republic       Portugal 2008          1 -0.1346162  -0.8883897       -3       -9258        -4
## 6     3       Portugal Czech Republic 2008          1  0.1346162   0.8883897        3        9258         4
```

```
head(Euro_test)
```

```
##   Goals        Team    Opponent Year GroupStage         GDP MarketValue FifaRank UefaPoints  CLPlayers
## 1     5     Germany    Scotland 2024          0  0.05468697   1.4117628      -23      50119  8.8461538
## 2     1    Scotland     Germany 2024          0 -0.05468697  -1.4117628       23     -50119 -8.8461538
## 3     1     Hungary Switzerland 2024          0 -1.62357231  -0.5393005        7     -11100 -0.8846154
## 4     3 Switzerland     Hungary 2024          0  1.62357231   0.5393005       -7      11100  0.8846154
## 5     3       Spain     Croatia 2024          0  0.46871918   1.0805475       -2      63714  2.6538462
## 6     0     Croatia       Spain 2024          0 -0.46871918  -1.0805475        2     -63714 -2.6538462
```

UEFA Euro Data

- The variables MarketValue and GDP are preprocessed with an intermediate step.
- **Do you have an idea why we cannot use these variables as they are?**

UEFA Euro Data

- The variables MarketValue and GDP are preprocessed with an intermediate step.
- **Do you have an idea why we cannot use these variables as they are?**
- GDP and MarketValue are highly depending on the respective year, and are not comparable across years.
  - The market values of players have risen dramatically since 2008.
  - The GDP of countries typically increase year by year.
- Therefore, these values are used log-transformed before we compute the respective differences between team and opponent.
- This way, the difference represents the log of the ratio of both values, which eliminates the effective size of the variables.

ATP Tennis Data

- The ATP Tennis data contain infos on grand slam matches of males in the years 2011–2023.
- Beside the names of the players and the result of the match, the data contain
  - differences of both players regarding age, ATP rank, ATP points, ELO rating
  - information on betting odds
  - the surface the respective tournament is played on
- Age is measured as the absolute difference from the "optimal" tennis age of 30 years.

ATP Tennis Data

```
head(tennis_slams)
```

```
##                 player.1         player.2 year      tournament surface win sets.1 sets.2 rank points
## 1           Adam Pavlasek Ernesto Escobedo 2017       Wimbledon   Grass   1      3      1   62   -275
## 2             Mirza Basic     Adam Pavlasek 2018     French Open    Clay   0      1      3 -104    400
## 3 Roberto Carballes Baena     Adam Pavlasek 2016     French Open    Clay   0      2      3  -18     81
## 4             Adam Walton   Federico Coria 2024       Wimbledon   Grass   1      3      0   29   -169
## 5    Alexander Shevchenko Adrian Mannarino 2023       Wimbledon   Grass   0      0      3   62   -489
## 6          Aslan Karatsev Adrian Mannarino 2022 Australian Open    Hard   0      1      3  -54   1674
##          elo       prob    age.30 AvgProb.1 AvgProb.2 B365.1 B365.2
## 1 -149.30944 -0.3395349 -1.736986 0.3302326 0.6697674   2.75   1.44
## 2   67.57861 -0.2801932 -3.241096 0.3599034 0.6400966   2.62   1.50
## 3   47.92105  0.1785714 -1.545812 0.5892857 0.4107143   1.53   2.37
## 4  -20.07617  0.5210728  2.478142 0.7605364 0.2394636   1.25   4.00
## 5  -55.55264 -0.5693950  2.389041 0.2153025 0.7846975   4.50   1.20
## 6  209.07964  0.5276190 -1.946545 0.7638095 0.2361905   1.25   4.00
```

Linear Regression

Linear Regression

- **Linear regression** is a method to describe the relationship between two or more variables.
- We distinguish between the **predictor variables** and the **response variable**.
- We always consider only a single response variable.
- In **simple linear regression** we have only a single predictor variable, in **multiple linear regression** we have multiple predictor variables simultaneously.
- In linear regression, we assume a **normal distribution** for the response.
- For the model of **simple linear regression**, we assume

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \,,$$

where $\beta_0 + \beta_1 x_i$ is known as the **systematic component** and ϵ_i is the random **error term**.

Simple Linear Regression

Simple Linear Regression

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \ , \quad i = 1, \ldots, n$$

$y_1, \ldots, y_n$ observations of response variable (on metric scale)

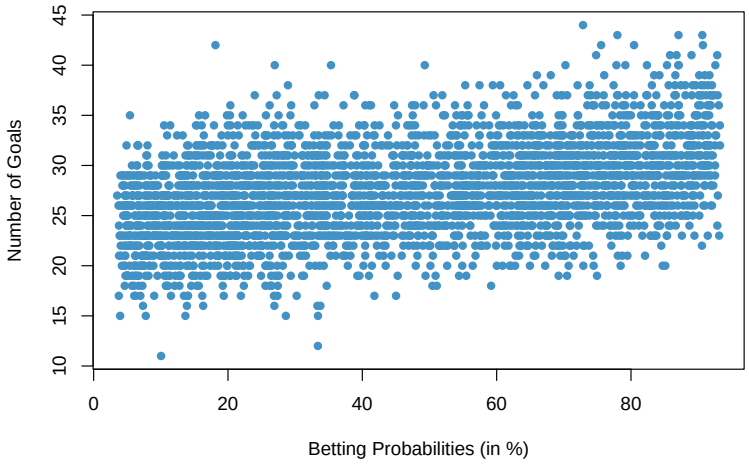
$x_1, \ldots, x_n$ observations of predictor variable

$\epsilon_1, \ldots, \epsilon_n$ random error term (residuals)

β_0 (unkown) intercept (has to be estimated!)

β_1 (unknown) regression coefficient (has to be estimated!)

Example: Handball Data

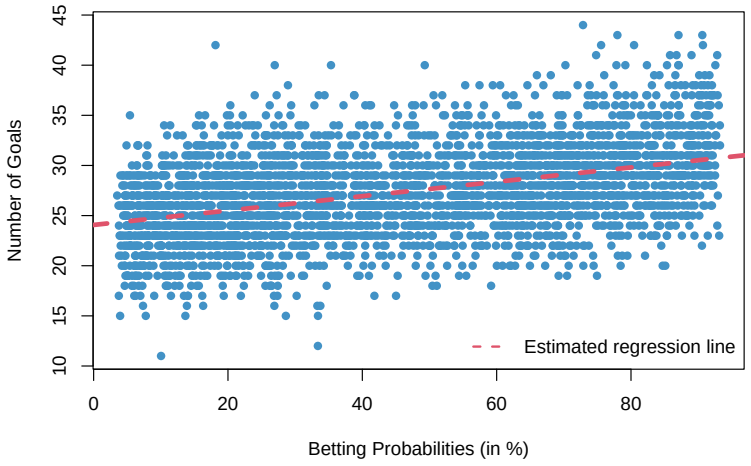


Example: Handball Data

```
lm.hb1 <- lm(goals ~ bet.probs, data = handball_long)
summary(lm.hb1)

##
## Call:
## lm(formula = goals ~ bet.probs, data = handball_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.4622  -2.6630  -0.1819   2.5125  16.6307
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 24.068837   0.132270   182.0   <2e-16 ***
## bet.probs    0.071688   0.002506    28.6   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.949 on 3514 degrees of freedom
## Multiple R-squared:  0.1889,Adjusted R-squared:  0.1886
## F-statistic: 818.2 on 1 and 3514 DF,  p-value: < 2.2e-16
```

Example: Handball Data



Example: Handball Data

Interpretation of estimated parameters:

$\hat{\beta}_0 = 24.068837$ (Estimated) intercept

→ No meaningful interpretation here! Represents the predicted number of goals for a team with a betting probability of 0%.

$\hat{\beta}_1 = 0.071688$ (Estimated) regression coefficient for betting probability

→ If the betting probability of a team increases by one percentage point, the number of goals increases (on average) by 0.072!

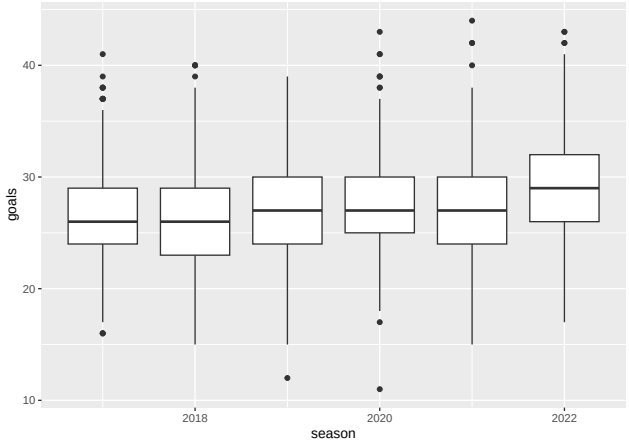
Extension to Multiple Linear Regression

- In many cases, it is important to consider more than one predictor variable simultaneously.
  - Sometimes, we consider further variables as they might also be powerful predictors for the response.
  - Sometimes, we consider further variables as they might be confounders for the effect of interest.
- In the handball data, we additionally consider whether the number of goals increases across seasons.

$\Rightarrow$ We can extend the simple linear model to the multiple linear model if we include more than one predictor variable!

Extension to Multiple Linear Regression

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The number of goals seems to (slightly) increase across seasons!

Multiple Linear Regression

Multiple Linear Regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip} + \epsilon_i \ , \quad i = 1, \ldots, n$$

$y_1, \ldots, y_n$	observations of response variable (metric)
$x_{11}, \ldots, x_{n1}$	observations of predictor variable 1
$\vdots$	$\vdots$
$x_{1p}, \ldots, x_{np}$	observations of predictor variable p
$\epsilon_1, \ldots, \epsilon_n$	random error term (residuals)
β_0	(unkown) intercept (has to be estimated!)
$\beta_1, \ldots, \beta_p$	(unknown) regression coefficients (have to be estimated!)

Example: Handball Data

```
lm.hb2 <- lm(goals ~ bet.probs + season, data = handball_long)
summary(lm.hb2)

##
## Call:
## lm(formula = goals ~ bet.probs + season, data = handball_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.2046  -2.5454  -0.1557   2.4930  15.4860
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.208e+02  7.720e+01  -11.93   <2e-16 ***
## bet.probs    7.177e-02  2.455e-03   29.24   <2e-16 ***
## season       4.679e-01  3.822e-02   12.24   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.868 on 3513 degrees of freedom
## Multiple R-squared:  0.2221,Adjusted R-squared:  0.2216
## F-statistic: 501.4 on 2 and 3513 DF,  p-value: < 2.2e-16
```

Example: Handball Data

Interpretation of estimated parameters:

$\hat{\beta}_0 = -920.8$ (estimated) intercept

→ Here, no meaningful interpretation! Represents the predicted number of goals for a team in season 0 with a betting probability of 0%.

$\hat{\beta}_1 = 0.07177$ (estimated) regression coefficient for betting probability

→ If the betting probability of a team increases by one percentage point (and season is constant), the number of goals increases (on average) by 0.072!

$\hat{\beta}_2 = 0.4679$ (estimated) regression coefficient for season variable

→ If the season increases by one year (and the betting probability is constant), the number of goals increases (on average) by 0.47!

Binary Predictors

- A variable which can only take two different values is called **binary** (e.g. home team with categories *home* and *away*).
- To include such a variable into a multiple linear model, we define a so-called dummy variable which takes only the values 0 or 1.
- Example:
  - In the handball data, the variable `home_away` contains the information whether the team is the home or the away team.
  - A home team is coded by 1 and an away team is coded by 0.
- The `lm(·)` function in `R` does this for us automatically!

Example: Handball Data

```
lm.hb3 <- lm(goals ~ bet.probs + season + home_away, data = handball_long)
summary(lm.hb3)

##
## Call:
## lm(formula = goals ~ bet.probs + season + home_away, data = handball_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.1878  -2.5485  -0.1551   2.4906  15.4995
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -9.208e+02  7.721e+01 -11.927   <2e-16 ***
## bet.probs      7.156e-02  2.566e-03  27.884   <2e-16 ***
## season         4.679e-01  3.823e-02  12.238   <2e-16 ***
## home_awayhome  3.879e-02  1.364e-01   0.284    0.776
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.868 on 3512 degrees of freedom
## Multiple R-squared:  0.2221,Adjusted R-squared:  0.2214
## F-statistic: 334.2 on 3 and 3512 DF,  p-value: < 2.2e-16
```

Example: Handball Data

- The parameter for the home/away effect is only 0.03879, i.e. the difference between home and away teams is (on average) only 0.04 goals.
- **Why is this effect so small?**
- **Is there no home effect in handball?**

Example: Handball Data

- The parameter for the home/away effect is only 0.03879, i.e. the difference between home and away teams is (on average) only 0.04 goals.
- **Why is this effect so small?**
- **Is there no home effect in handball?**

$\Rightarrow$ The betting probabilities already cover the home effect!

Example: Handball Data

Interpretation of estimated parameters:

$\hat{\beta}_0 = -902.8$ (estimated) intercept

$\hat{\beta}_1 = 0.07156$ (estimated) regression coefficient for betting probability

→ If the betting probability of a team increases by one percentage point (and season and home_away are constant), the number of goals increases (on average) by 0.072!

$\hat{\beta}_2 = 0.4679$ (estimated) regression coefficient for season variable

→ If the season increases by one year (and the betting probability and home_away are constant), the number of goals increases (on average) by 0.47!

$\hat{\beta}_3 = 0.03879$ (estimated) regression coefficient for home variable

→ For equal values of the betting probability and season, home teams score (on average) 0.039 goals more than the away team!

Coefficient of Determination

- An important statistic to check the goodness-of-fit is the **coefficient of determination** R^2.
- R^2 expresses the percentage of variance of the response that is explained by the regression.
- Possible values: $0 \leqslant R^2 \leqslant 1$
- Larger values of R^2 refer to a better fit of the model.
- On Slide 31 we can find the R^2 for our latest model under

```
Multiple R-squared:  0.2221
```

$\Rightarrow$ 22.2% of the variance of the number of goals can be explained by our model!

Example: Handball Data

- For comparison, we also try to model the goal difference.
- For this, we need the data set `handball_wide`, using bet.probs.diff and season as predictors and `scorediff` as the match outcome.
- Principally, we can use R^2 as a simple tool of comparison between both models: the model with the larger R^2 value fits the data better!
- However, note that it is not possible to directly compare simpler to more complex models based on the standard R^2; if the simpler model is based on covariates that are a subset of those of the more complex model, then the more complex model will always yield the larger R^2.

  $\rightarrow$ For this purpose, the **adjusted** R^2 could be used.

Example: Handball Data

```
lm.hb4 <- lm(scorediff ~ bet.probs.diff + season, data = handball_wide)
summary(lm.hb4)

##
## Call:
## lm(formula = scorediff ~ bet.probs.diff + season, data = handball_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.1625  -2.8487  -0.0452   2.8036  20.3266
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -87.13772  125.21213  -0.696    0.487
## bet.probs.diff   7.11107    0.20836  34.129   <2e-16 ***
## season           0.04317    0.06200   0.696    0.486
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.435 on 1755 degrees of freedom
## Multiple R-squared:  0.3989,Adjusted R-squared:  0.3982
## F-statistic: 582.4 on 2 and 1755 DF,  p-value: < 2.2e-16
```

Example: Handball Data

- The model using score differences has a much better fit compared to the model for the absolute number of goals.
- Compared to the model for the absolute number of goals ($R^2 = 22\%$), the model for the goal difference can explain around 40% of the variance in the outcome variable!
- **What could be the reason why this model fits so much better?**

Example: Handball Data

- The model using score differences has a much better fit compared to the model for the absolute number of goals.
- Compared to the model for the absolute number of goals ($R^2 = 22\%$), the model for the goal difference can explain around 40% of the variance in the outcome variable!
- **What could be the reason why this model fits so much better?**
  - In handball, there are often certain dynamics with repsect to the pace of the game, leading to either many or only few goals (for both teams).
  - Second, the teams might only be focused on the goal difference and not the absolute number and might be less engaged to keep on scoring once the match appears to be decided.

Hypothesis Testing in Linear Regression

- The standard R output for a linear regression model contains the results of two different types of tests, which are the **overall F-test** and individual tests separately for each coefficient. We spare the mathematical details here.
- The overall F-test is used to test all parameters simultaneously. It has the following hypotheses:

$$H_0: \ \beta_1 = \ldots = \beta_p = 0 \quad \text{vs.} \quad H_1: \ \beta_j \neq 0 \text{ for at least one } j$$

  Per regression model, only one p-value for the overall F-test exists.
- Additionally, each individual regression coefficient β_j can be tested:

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0$$

  For each regression coefficient, a separate p-value is displayed in the R output.

Example: Handball Data

```
lm.hb3 <- lm(goals ~ bet.probs + season + home_away, data = handball_long)
summary(lm.hb3)

##
## Call:
## lm(formula = goals ~ bet.probs + season + home_away, data = handball_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.1878  -2.5485  -0.1551   2.4906  15.4995
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -9.208e+02  7.721e+01 -11.927   <2e-16 ***
## bet.probs      7.156e-02  2.566e-03  27.884   <2e-16 ***
## season         4.679e-01  3.823e-02  12.238   <2e-16 ***
## home_awayhome  3.879e-02  1.364e-01   0.284    0.776
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.868 on 3512 degrees of freedom
## Multiple R-squared:  0.2221,Adjusted R-squared:  0.2214
## F-statistic: 334.2 on 3 and 3512 DF,  p-value: < 2.2e-16
```

Example: Handball Data

- On the previous slide we again saw the output of the regression model with predictors `bet.probs`, `season`, and `home_away`.
- The p-value for the overall F-test ($< 2.2e-16$) is found in the last line.

$\Rightarrow$ Highly significant!

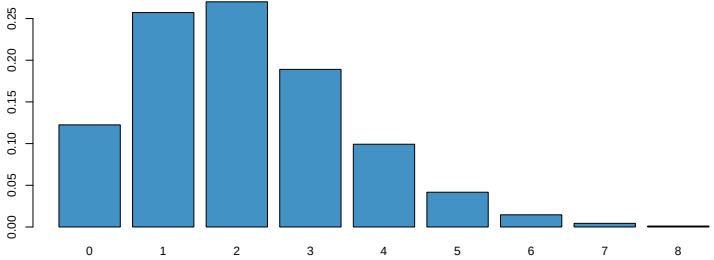
- The p-values for the individual tests are found in the last column of the coefficients matrix.

$\Rightarrow$ bet.probs and season are highly significant, home_away is not significant!

Poisson Regression

Poisson Regression

- In **Poisson regression**, we assume that the outcome follows a Poisson distribution: $y_i \sim \mathcal{P}(\lambda_i)$.
  - A Poisson distribution is used to model the number of events in a certain time frame.
  - In football, the parameter λ_i represents the expected number of goals per match.
- Example: distribution of number of goals for $\lambda_i = 2.1$



Poisson Regression

- Poisson regression connects the rate $\lambda_i = \mathrm{E}[y_i]$ of the Poisson distribution with the **linear predictor**

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip}$$

  via

$$\lambda_i = \exp(\eta_i) = \exp(\beta_0) \exp(\beta_1 x_{i1}) \cdot \ldots \cdot \exp(\beta_p x_{ip})$$

- The exponential function $\exp(\cdot)$ serves as the **response function**, which connects the linear predictor η_i to the outcome and guarantees $\lambda_i > 0$. In linear regression, the response function is simply the identity function.

- The effect of covariates on the rate λ_i is **exponentially multiplicative**. This means that an increase in x_{ij} by 1 unit leads to multiplication of the expected events (i.e. expected number of goals) by the factor $exp(\beta_j)$.

Example: UEFA Euro Data

- We use the UEFA Euro Data as an exemplary data set to illustrate Poisson regression.
- In R, we use the `glm()` function to fit so-called Generalized Linear Regression Models (GLMs), where Poisson and Logistic Regression (see next topic) are the most prominent examples.
- GLMs extend linear regression to different outcome distributions beyond the normal distribution.

Example: UEFA Euro Data

```
pois.euro1 <- glm(Goals ~ GroupStage + GDP + MarketValue + FifaRank + UefaPoints + CLPlayers,
                  family = poisson(), data = Euro_train)
summary(pois.euro1)

##
## Call:
## glm(formula = Goals ~ GroupStage + GDP + MarketValue + FifaRank +
##     UefaPoints + CLPlayers, family = poisson(), data = Euro_train)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.604e-01  9.034e-02   1.775   0.0759 .
## GroupStage  -4.925e-02  1.046e-01  -0.471   0.6377
## GDP          1.011e-01  4.427e-02   2.283   0.0224 *
## MarketValue  1.742e-01  8.326e-02   2.092   0.0365 *
## FifaRank    -2.532e-03  2.990e-03  -0.847   0.3970
## UefaPoints  -7.768e-07  2.338e-06  -0.332   0.7397
## CLPlayers    2.672e-02  1.685e-02   1.586   0.1128
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 470.63  on 389  degrees of freedom
## Residual deviance: 412.89  on 383  degrees of freedom
## AIC: 1068.5
##
## Number of Fisher Scoring iterations: 5
```

Exemplary Interpretation of Estimated Parameters

$\hat{\beta}_3 = 0.1742$ (Estimated) regression coefficient for MarketValue

→ $\exp(0.1742) = 1.190257$

→ If the difference between the log market values of the competing teams increases by 1 and all other variables are constant, the expected number of goals of a team grows by the factor 1.19 (i.e. by 19%)!

- For an increasing FIFA rank (increasing means worse!) the expected number of goals decreases; for an increasing GDP and an increasing number of CL players the expected number of goals increases (however, not all of these effects being significant).

→ These effects are intuitive!

- For an increasing number of UEFA points, the expected number of goals is decreasing.

→ This is counterintuitive! **Do you have an explanation for this?**

Exemplary Interpretation of Estimated Parameters

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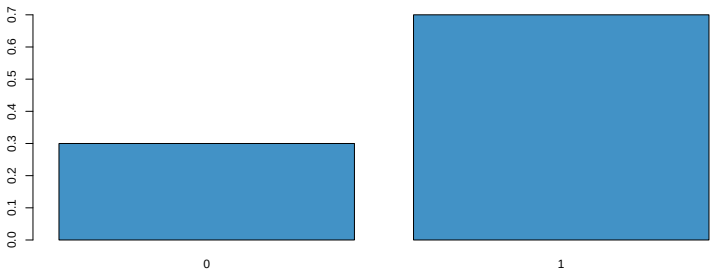
→ This is counterintuitive! **Do you have an explanation for this?**

→ The parameter is tiny! UEFA points seem to deliver no **additional** information beyond what is already explained by the other variables.

Logistic Regression

Logistic Regression

- In **Logistic regression**, we assume a Bernoulli distribution for the response: $y_i \sim \mathcal{B}(1, p_i)$
  - A Bernoulli distribution is used to model binary responses (e.g. player A wins or loses)
  - In tennis, the parameter p_i represents the probability of player A to win
- Example: Bernoulli distribution for $p_i = 0.7$



Logistic Regression

- Logistic regression connects the probability $p_i = \mathrm{E}[y_i]$ of the Bernoulli distribution with the **linear predictor**

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip}$$

  via the so-called logistic function

$$P(y_i = 1) = p_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}.$$

- The logistic function $\frac{\exp(\cdot)}{1+\exp(\cdot)}$ serves as the **response function** which connects the linear predictor η_i to the outcome.

- Here, the effect of covariates on the odds $\frac{p_i}{1-p_i}$ is **exponentially multiplicative**. This means that an increase in x_j by 1 unit leads to multiplication of the odds (e.g. the odds to win the match) by the factor $\exp(\beta_j)$.

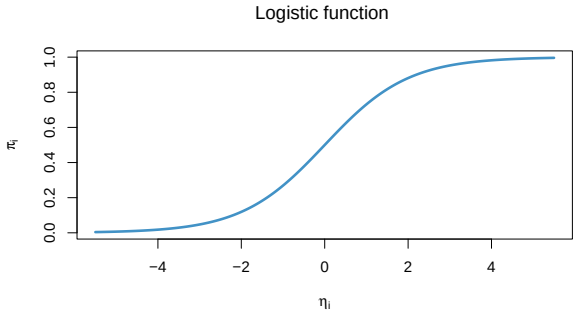
- Accordingly, the exponentials of the parameters from logistic regression (i.e. $\exp(\beta_j)$) are typically interpreted as (adjusted) odds ratios.

Logistic Regression

- Logi… the **linea…**

via t…

- The … r pred…
- Here… means that … the matc…
- Acco… e typic…



$$P(y_i = 1) = p_i = \frac{exp(\eta_i)}{1 + exp(\eta_i)}$$

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip}$$

Logistic Regression

- Logistic regression connects the probability $p_i = \mathrm{E}[y_i]$ of the Bernoulli distribution with the **linear predictor**

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip}$$

  via the so-called logistic function

$$P(y_i = 1) = p_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}.$$

- The logistic function $\frac{\exp(\cdot)}{1+\exp(\cdot)}$ serves as the **response function** which connects the linear predictor η_i to the outcome.

- Here, the effect of covariates on the odds $\frac{p_i}{1-p_i}$ is **exponentially multiplicative**. This means that an increase in x_j by 1 unit leads to multiplication of the odds (e.g. the odds to win the match) by the factor $\exp(\beta_j)$.

- Accordingly, the exponentials of the parameters from logistic regression (i.e. $\exp(\beta_j)$) are typically interpreted as (adjusted) odds ratios.

Example: Tennis data

```
logistic.tennis1 <- glm(win ~ rank + points + elo + age.30 - 1, data = tennis_slams, family = "binomial")
summary(logistic.tennis1)
```

```
##
## Call:
## glm(formula = win ~ rank + points + elo + age.30 - 1, family = "binomial",
##     data = tennis_slams)
##
## Coefficients:
##         Estimate Std. Error z value Pr(>|z|)
## rank   1.517e-04  3.676e-04   0.413  0.67995
## points 6.847e-05  2.147e-05   3.190  0.00143 **
## elo    5.870e-03  2.657e-04  22.095  < 2e-16 ***
## age.30 1.481e-02  7.832e-03   1.891  0.05858 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 9177.3  on 6620  degrees of freedom
## Residual deviance: 6673.5  on 6616  degrees of freedom
## AIC: 6681.5
##
## Number of Fisher Scoring iterations: 5
```

Example: Tennis data

```
logistic.tennis1 <- glm(win ~ rank + points + elo + age.30 - 1, data = tennis_slams, family = "binomial")
summary(logistic.tennis1)

##
## Call:
## glm(form
##     data
##
## Coeffici
##
## rank
## points (
## elo
## age.30
## ---
## Signif.
##
## (Dispers
##
##     Null
## Residual
## AIC: 668
##
## Number of Fisher Scoring iterations: 5
```

Hint:

- In the formula, we eliminate the intercept by adding -1
- The reason why we need no intercept is that there is no reason why the first and the second named player should have unequal odds of winning if all predictors are 0.
- This would only be true if, for example, the first-named player (or team) is always the home-player (or home team)
- However, in tournaments (in contrast to league data) we typically have no home advantage and being first- or second-named player is just a coincidence.

Exemplary Interpretation of Estimated Parameters

$\hat{\beta}_3 = 0.00587$ (Estimated) regression coefficient for Elo rating

→ $\exp(0.00587) = 1.005887$

→ If the difference between the Elo ratings of the competing players increases by 1 and all other variables are constant, the odds of winning grow by the factor 1.006 (i.e. by 0.6%)!

- For an increasing Elo rating and an increasing number of points, the odds of winning increase.
  → This effect is intuitive!
- For an increasing (i.e. worse) rank, the odds of winning increase.
  → This is counterintuitive, but again it seems that everything is already covered by the other predictors (the p-value for rank is the largest)

Example: Tennis data

Using the following steps, we try to evaluate how well we can predict the final results using our model

1. Use `predict()` to get winning probabilities for first-named player according to the model
2. Create predicted match result by assigning a win for probabilities $> 50\%$ and vice versa
3. Create 2×2 between actual and predicted binary match outcome

```
tennis_slams$pred.prob <- predict(logistic.tennis1, type = "response")
tennis_slams$y.pred <- ifelse(tennis_slams$pred.prob>0.5, 1, 0)
table(tennis_slams$y.pred, tennis_slams$win)

##
##        0    1
##   0 2465  824
##   1  839 2492
```

The majority of the matches (4957 out of 6620) is predicted correctly by the model!

Summary

Summary

- Match outcomes can be different types of variables and, therefore, require different types of models.
- The most frequently occurring outcomes are
  - Scores (two observations per match)
    - Linear regression (for large scores)
    - Poisson regression (for small scores)
  - Score differences (one observation per match)
    - Linear regression (for many possible difference values)
    - Skellam regression (for few possible difference values)
  - Binary outcomes → Logistic regression
- Interpretation of parameters in Poisson regression and Logistic regression is exponentially multiplicative (either on the expected number of scores or on the odds).